

TECHNICAL REPORT 003

Closed-Loop Physical-Design Search

Using routed observations to improve workload-specific accelerator choices
under a fixed implementation budget

RESEARCH LANE

CLOSED-LOOP ARCHITECTURE SEARCH

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K1 closed-loop research report

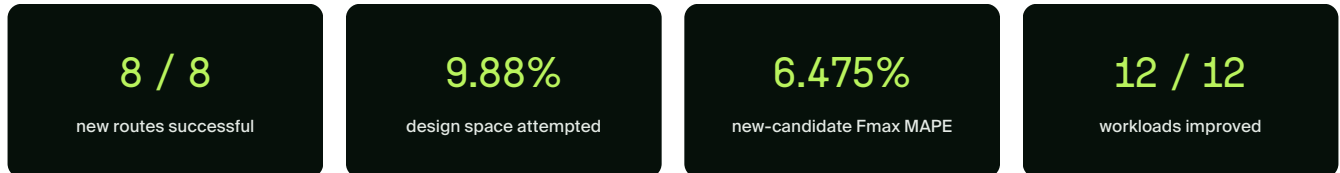
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ABSTRACT

Route a few designs, learn where to route next

Kernellum K1 tests whether physical-design observations can guide architecture search more effectively than an equal-budget random control. The frozen architecture pool contains 172 ECP5 candidates. Starting from nine routed observations, Kernellum chooses four new designs using a final-route-Fmax surrogate and an uncertainty-aware acquisition rule. The equal-budget random arm receives the same four-route budget.



Closed-loop result

Active search reduced mean final-best workload latency to **16.7398 ms** versus **19.4428 ms** for the equal-budget random arm and improved the routed optimum on all 12 Transformer GEMMs.

01 / BASELINE

Starting point and acquisition engine

The baseline K1 experiment routes nine frozen architectures and tests whether fixed-frequency analytical rankings remain useful after place-and-route. All **9/9** route successfully, mean workload-rank Spearman is **0.9441**, and mean predicted-winner routed regret is **0.5828%**.

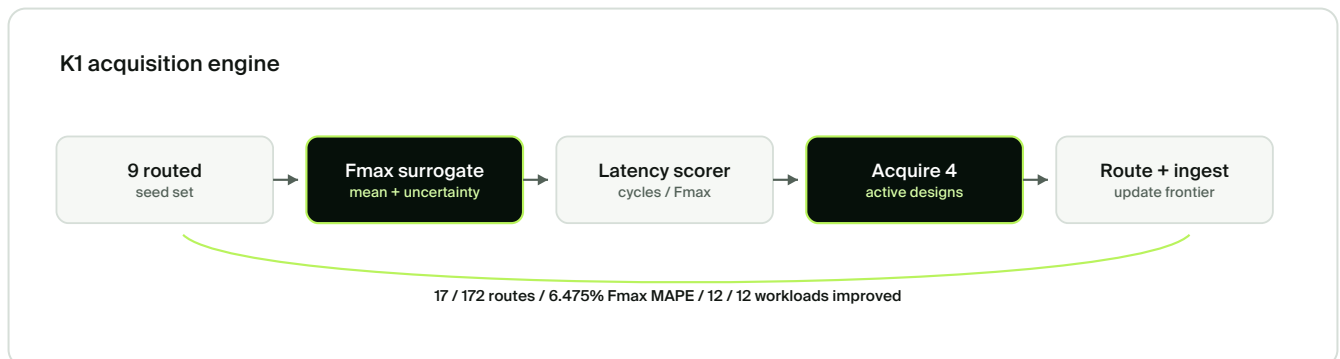


Figure 1. Closed-loop acquisition engine. The routed seed set fits a target-aware surrogate, scores the unobserved pool through a workload-latency objective, acquires four candidates, and ingests the new final-route measurements.

The complete program remains inside the frozen implementation budget: nine initial routes + four active routes + four random-control routes = **17/172** physically attempted architectures.

02 / ACTIVE VERSUS RANDOM

Acquired architectures

Arm	Architecture	Predicted Fmax	Final-route Fmax	DSP	FF
Active	10x12 K64	47.23	41.95	120	4036
Active	10x12 K32	47.84	45.37	120	3982
Active	12x10 K64	47.14	44.80	120	4035
Active	8x14 K64	47.76	45.77	112	3780
Random	2x14 K16	51.19	54.48	28	1004
Random	8x6 K8	49.69	53.42	48	1608
Random	4x6 K8	51.31	55.33	24	834
Random	8x4 K8	51.89	53.98	32	1096

The active selector deliberately explores larger workload-relevant arrays even when their absolute Fmax is below smaller random controls. The decision objective is workload latency, not standalone frequency.

03 / PREDICTION

Surrogate quality

The surrogate is used to decide where physical-design budget is likely to improve workload latency. Across the eight new candidates, final-route-Fmax MAPE is **6.475%**.

Decision quantity

For workload w and architecture a , K1 evaluates the routed-latency proxy

$$\hat{L}(w, a) = \frac{\text{cycle_count}(w, a)}{\hat{F}_{\text{route}}(a)}$$

The numerator is fixed by the kernel model; the denominator is the predicted final-route frequency. Acquisition prioritizes candidates that improve the workload frontier while covering uncertainty.

Surrogate prediction versus final-route Fmax

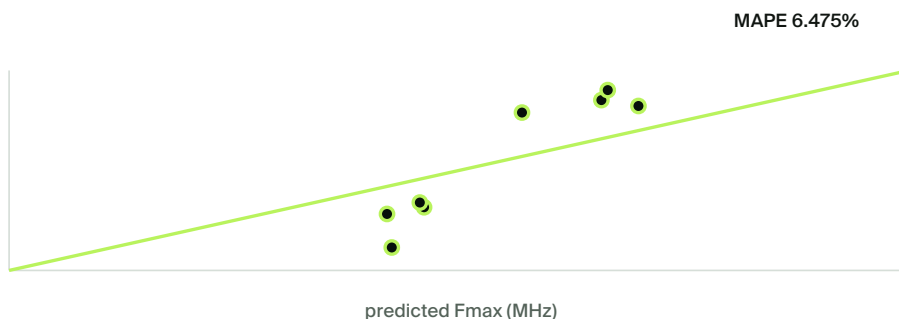


Figure 2. Surrogate prediction versus observed final-route Fmax for all eight new candidates. The diagonal is exact agreement.

04 / ROUTED OPTIMUM

Workload result

Representative K1 workload improvements

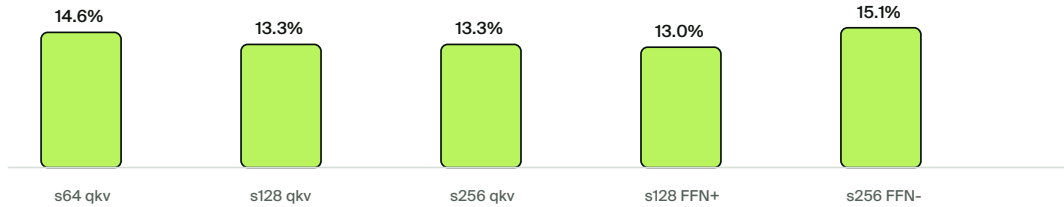


Figure 3. Representative workload improvements from the active final best over the original routed frontier.

Workload	Initial best	Active final best	Improvement
s64 qkv	8.790 ms	7.506 ms	14.6%
s128 qkv	16.918 ms	14.662 ms	13.3%
s256 qkv	33.835 ms	29.324 ms	13.3%
s128 FFN expand	21.597 ms	18.797 ms	13.0%
s256 FFN contract	44.177 ms	37.489 ms	15.1%

Across all 12 workloads, active mean final-best latency is **16.7398 ms** versus **19.4428 ms** for the equal-budget random arm, a **13.90%** reduction under the same four-route acquisition budget.

05 / EVIDENCE

Validation gate and mechanism

Metric	Required	Observed
New routes successful	$\geq 7 / 8$	8 / 8
Physically attempted fraction	$\leq 10\%$	9.88%
Surrogate Fmax MAPE	$\leq 20\%$	6.475%
Active mean latency vs random	no worse	16.74 vs 19.44 ms
Active workload improvements	$\geq$ random	12 vs 0

K1 establishes that final-route feedback can improve architecture search. Project SIMILARITY then isolates a specific physical mechanism that helps explain why choices move after routing: operand-distribution topology.

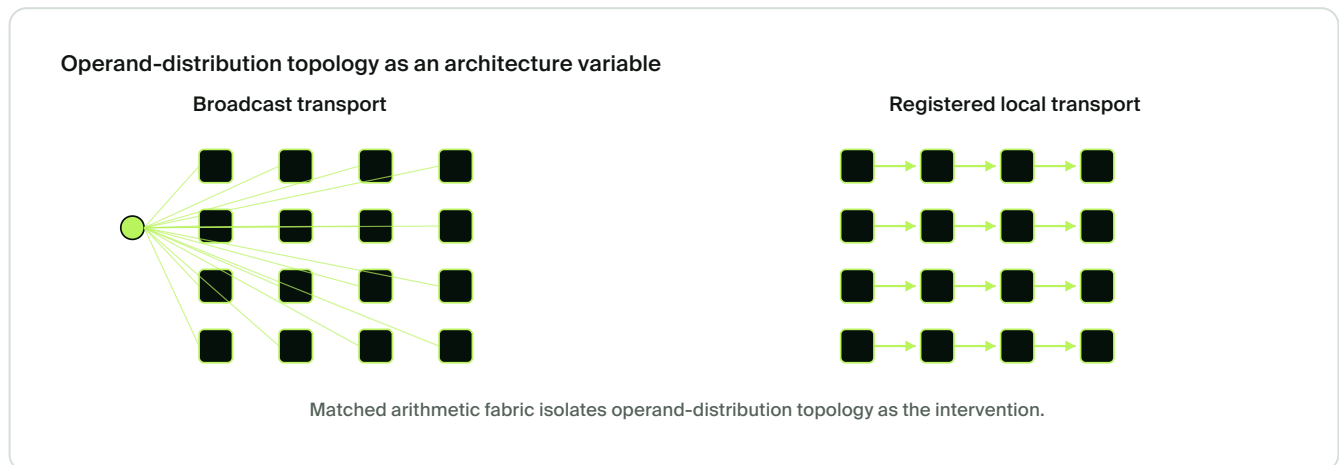


Figure 4. Mechanism bridge to SIMILARITY. Broadcast fanout grows more costly with array size than registered local transport in the tested ECP5 family.

O6 / HARDWARE MEASUREMENT

Bridge to K2

Three K2 board images are already built for the ECP5 EVN stack: baseline 8x8 K32, active 10x12 K32, and active 8x14 K64. The images pass simulation, synthesis, place-and-route, a 12 MHz timing target, and bitstream generation.

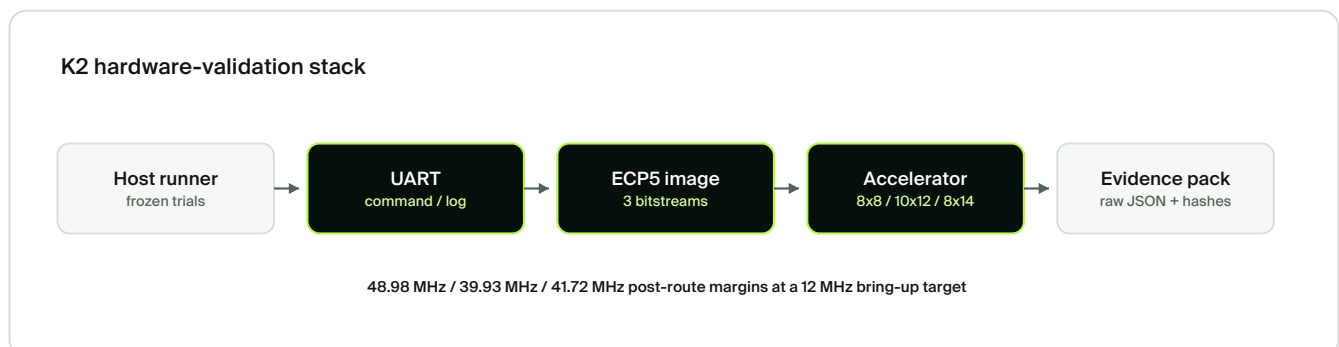


Figure 5. K2 measurement pathway. The closed-loop routed-search program hands off into a frozen host-to-board validation stack with raw hardware logs and bitstream hashes.

Publication record

Kernellum Research Technical Report 003. Primary records: results/k1_closed_loop_proposals.json, results/k1_closed_loop_routes.csv, and results/k1_closed_loop_validation.json.